

## M002 Week 6 — Redstone & Levers

**Topic:** Redstone Levers · **Course:** Maze Madness · **Level:** Advanced · **Time:** about 55 minutes

The maze has **redstone levers**. The agent detects redstone, flips a lever with `interact` to open the door, then walks through to clear the maze.

### 1 · The API

| Code                                         | What it does              |
|----------------------------------------------|---------------------------|
| <code>agent.detect(REDSTONE, FORWARD)</code> | True if redstone is ahead |
| <code>agent.interact(...)</code>             | flip the lever in front   |
| <code>agent.move(FORWARD, 1)</code>          | step one block            |
| <code>agent.turn(RIGHT_TURN)</code>          | turn right                |
| <code>player.on_chat("go", run)</code>       | run code on chat word     |

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## 2 · Predict 🧠

```
agent.move(FORWARD, 1)
agent.interact(...)
agent.move(FORWARD, 1)
```

**The agent reaches a lever and flips it. What happens to the door?**

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```
agent.interact(...)
agent.move(FORWARD, 1)
agent.move(FORWARD, 1)
```

**There is no lever in front. What happens?**

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### 3 · Trace the Code 🔍

```
while not agent.detect(REDSTONE, FORWARD):  
    agent.move(FORWARD, 1)  
agent.interact(...)  
agent.turn(RIGHT_TURN)  
agent.move(FORWARD, 1)  
agent.move(FORWARD, 1)
```

**Describe what the agent does, step by step, in plain English.**

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## 4 · Spot and Fix the Bugs 🐛

Write the fixed code, then say why.

**Bug A** — Flip the lever **first**, then walk through the open door. The lever is at the first wall.

```
agent.move(FORWARD, 1)
agent.move(FORWARD, 1)
agent.interact(...)
```

**Fixed code:**

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**Why was it wrong?**

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**Bug B** — Walk **up to** the lever, then flip it **once**. This flips on every step.

```
while not agent.detect(REDSTONE, FORWARD):
    agent.move(FORWARD, 1)
    agent.interact(...)
```

**Fixed code:**

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**Why was it wrong?**

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**Bug C** — After flipping, the agent must **turn around** and go back through the door. The door is behind it.

```
agent.interact(...)
agent.move(FORWARD, 1)
```

**Fixed code:**

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**Why was it wrong?**

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## 5 · Write the Code 🖋️

Somewhere ahead is a lever. Write code that walks to it, flips it, and walks through the open door.

```
def run():
    # your code here

player.on_chat("go", run)
```

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## 6 · Show Your Work 📷 🎥

Open your homework world. The maze has a **lever** that opens a **closed door**. Write code that:

1. moves the agent to the lever,
2. flips it with `interact` ,
3. walks through the open door,
4. reaches the goal.

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

**1 · Your code.** Scroll through it. Say what each part does.

**2 · Your build.** Point the camera. Name the parts.

Fill the blanks:

Today I built \_\_\_\_\_.

I built it using this code: \_\_\_\_\_.

In this code I used \_\_\_\_\_.

The hardest part was \_\_\_\_\_.

That part was hard because \_\_\_\_\_.

The most fun part was \_\_\_\_\_.

Something new I learned was \_\_\_\_\_.

**Write your lines here, then say them in your video.**

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**Submit** 

Send this worksheet + your video to teacher on KakaoTalk.